



Chapter 7 Iteration

CS 101- Programming
Fundamentals

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For loop

- Iterate over a sequence (list, tuple, string) or other iterable objects.
- Iterating over a sequence is called **traversal**.
- Flow of execution runs through the body and then *loops back* to the top
- Syntax:
 - **header** that ends with a colon and an indented body.
 - body can contain any number of statements
 - Optional: *else* block

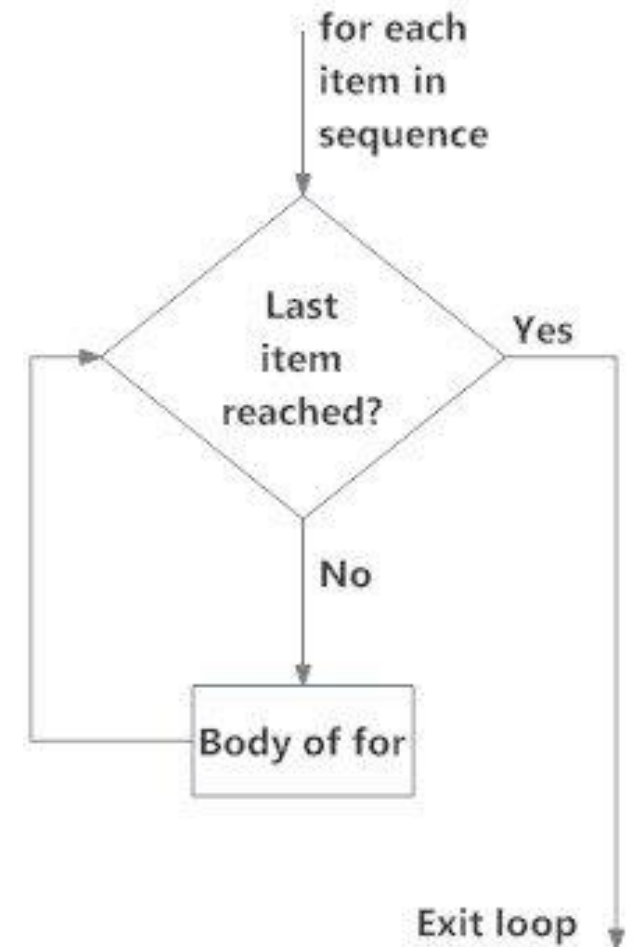


Fig: operation of for loop

Range ()

- Generates a sequence of numbers:
 - `range (4)` = Go from 0 to 3
- If start, stop and step size specified:
 - `range(start, stop, step_size)`
 - `range(10, 1, -2)` = Go from 10 to 1, by decrementing 2.
- To-do: Print the range 0, 5, 10, 15, 20, 25 with iteration

Break, continue and else with loop

- **Break:**
 - Check condition anywhere in the loop
 - If true, jump out of the loop.
- **Continue:** End the current iteration in a loop, and continue to the next iteration.
- **Else:** Executed if the items in the sequence used in for loop exhausts.

While statement

- Determine whether the condition is true or false.
- If false, exit the while statement and continue execution at the next statement.
- If the condition is true, run the body and then go back to step 1

Syntax: `while test_condition:`
 `Statement(s)`